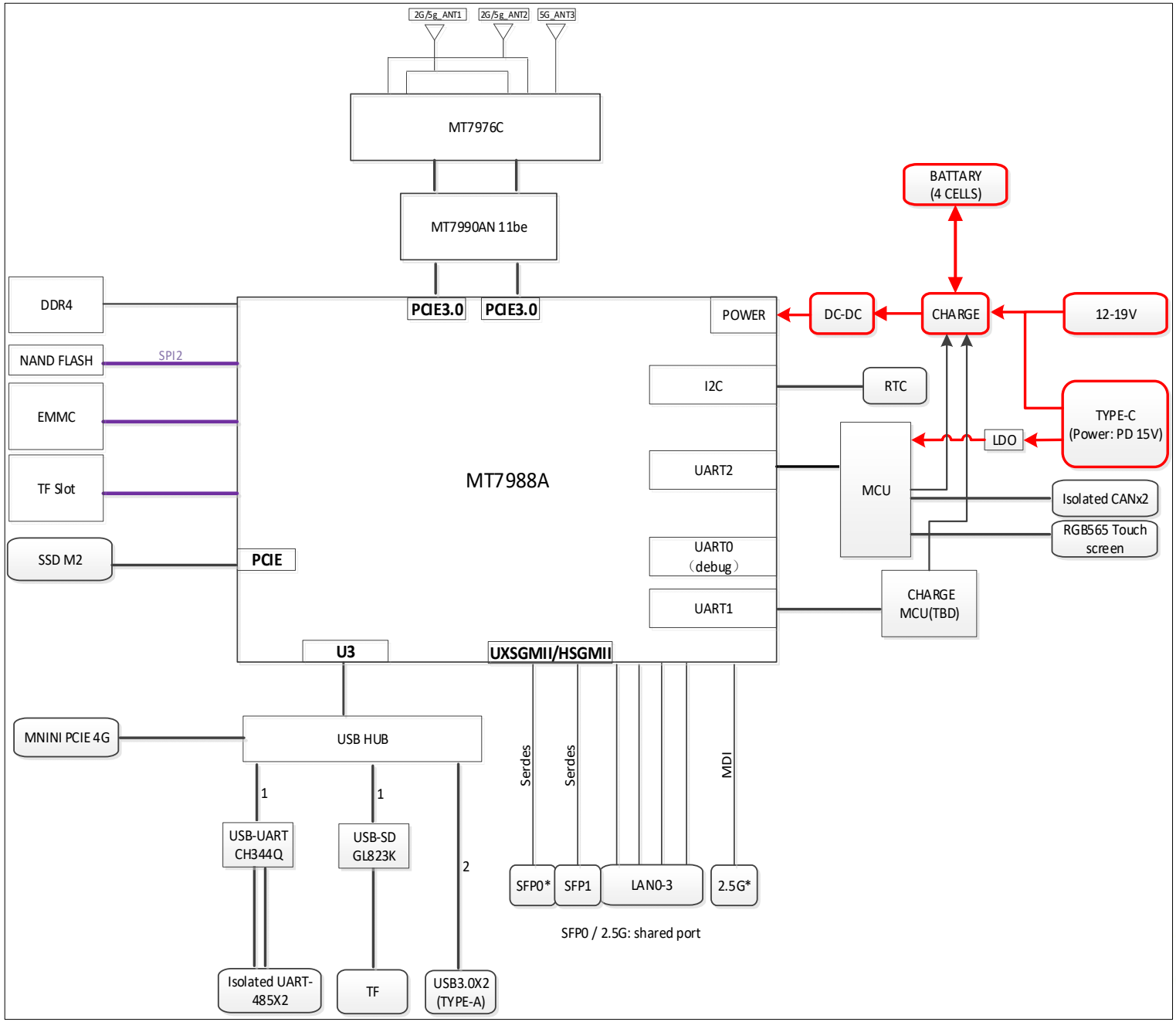
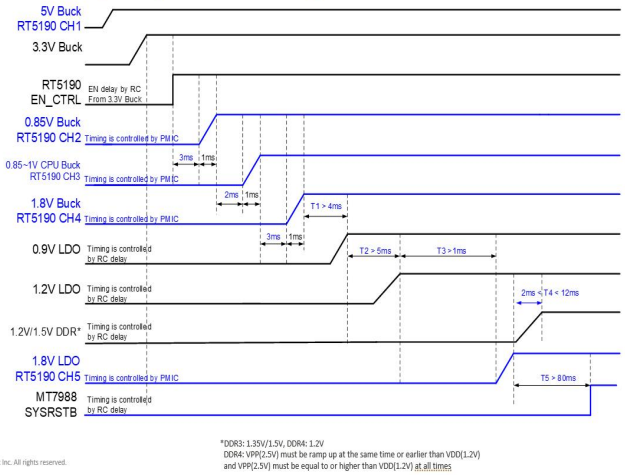
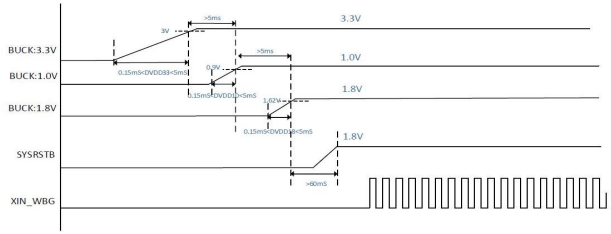
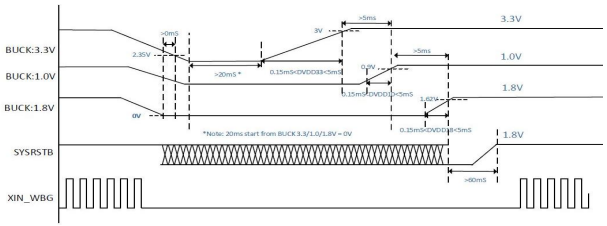


POWER:
MT7988A+PCIe 2Lx2+ USXGMIlx1 +1Gx4 + USBx2 + int.2.5G x1:MAX 7.5W
MT7990:7.87W
MT7976C:1.99W

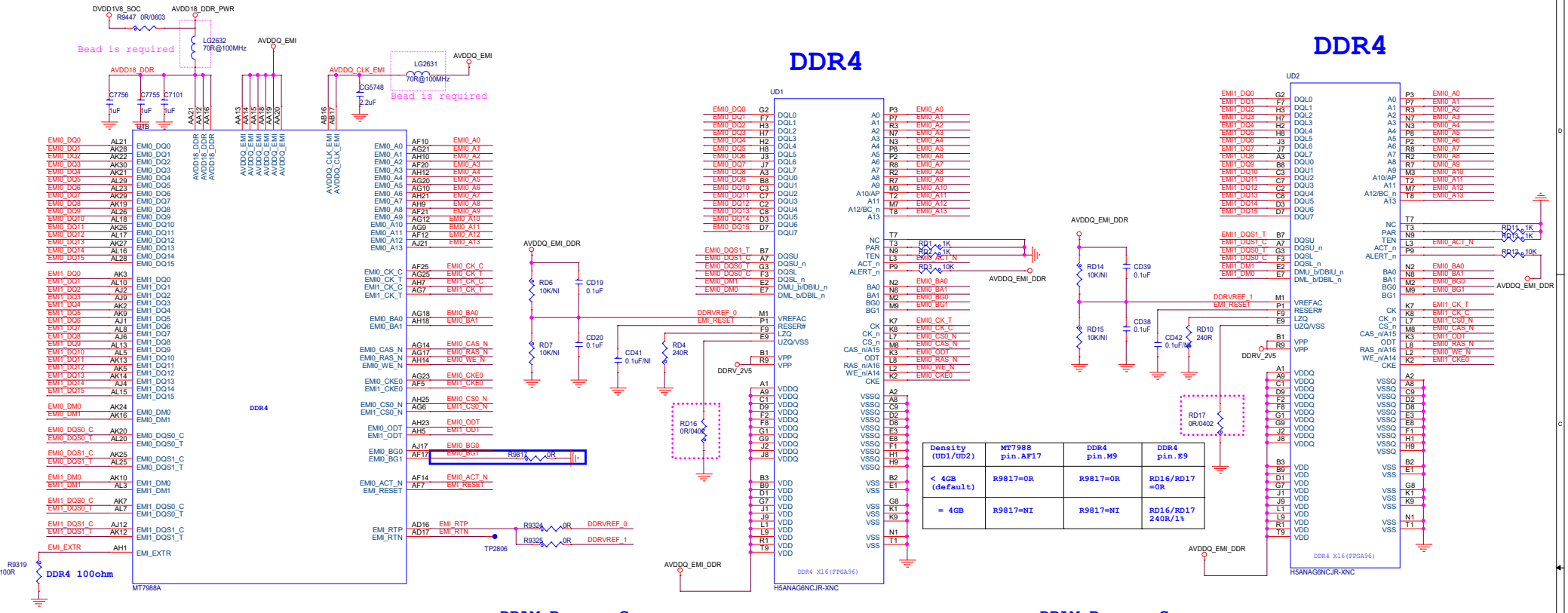
Jaguar power on sequence @20220627





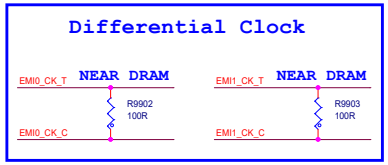


Title: Block Diagram			
Size: A1	Document Number: U7988A-09	Drawn:	Rev: V1.0
Date: Tuesday, May 26, 2026	Sheet 1 of 19		

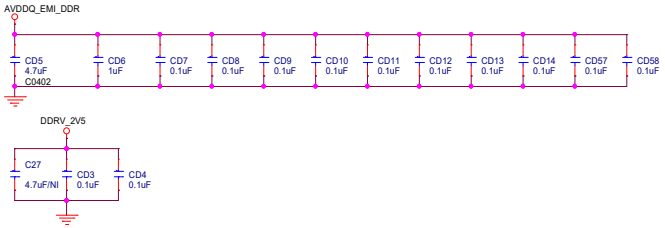


Density (UD1/UD2)	MT7988 pin.AF17	DDR4 pin.M9	DDR4 pin.E9
< 4GB (default)	R9817=0R	R9817=0R	RD16/RD17 =0R
= 4GB	R9817=NI	R9817=NI	RD16/RD17 240R/1%

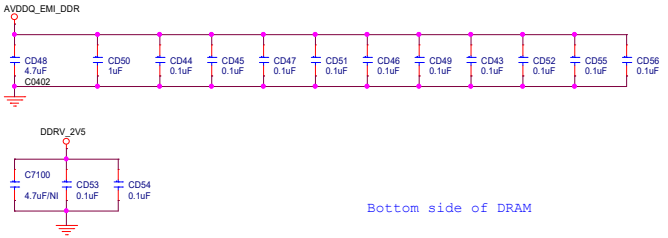
Differential Clock



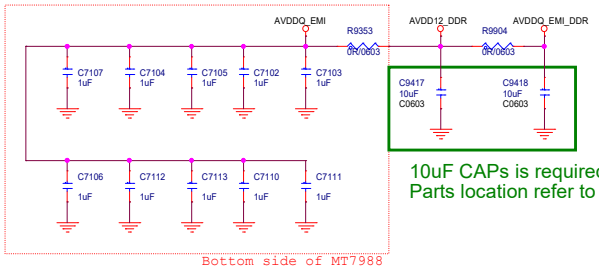
DRAM Bypass Cap



DRAM Bypass Cap



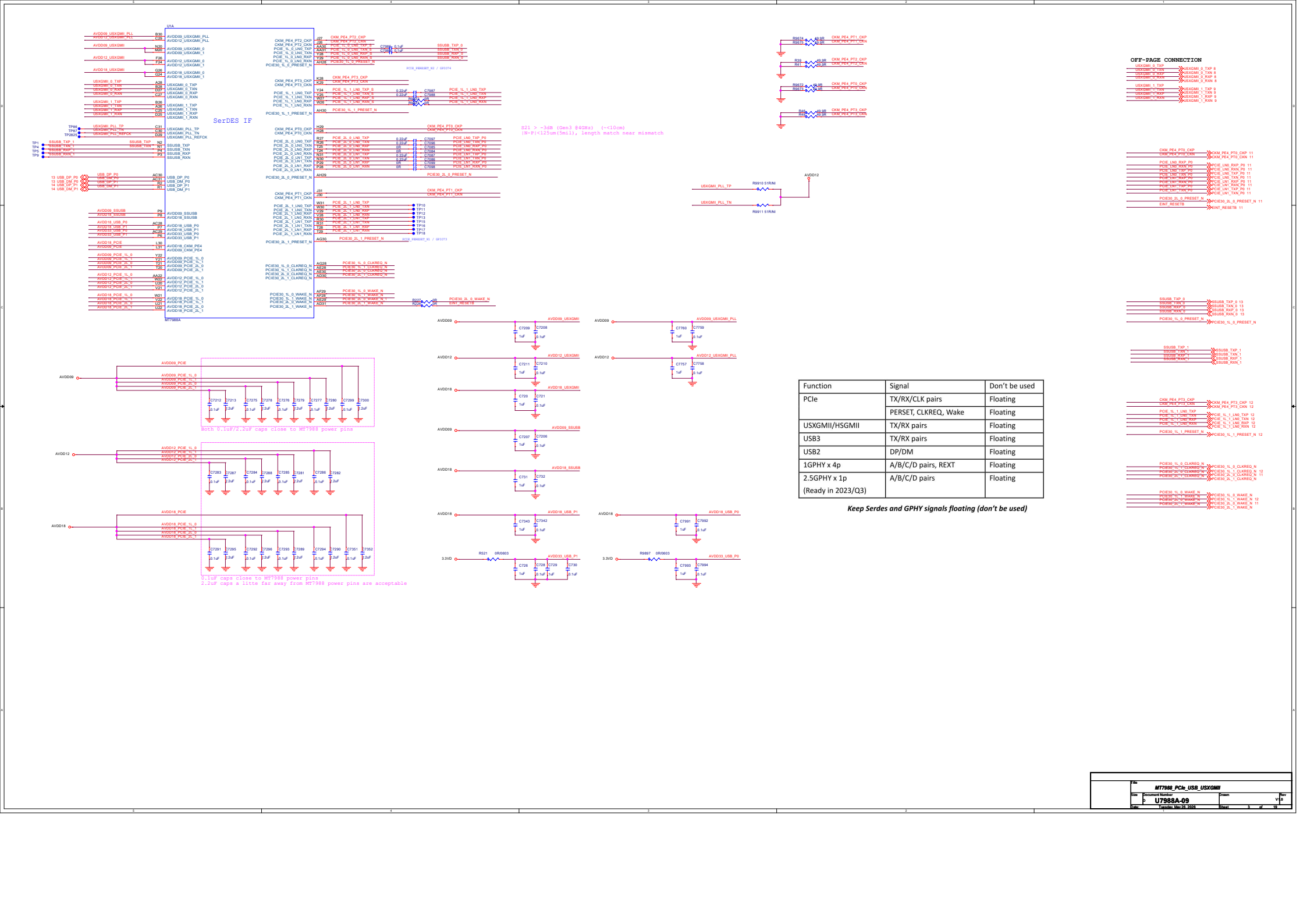
Bottom side of DRAM



Bottom side of MT7988

10uF CAPs is required.
Parts location refer to HDK layout.

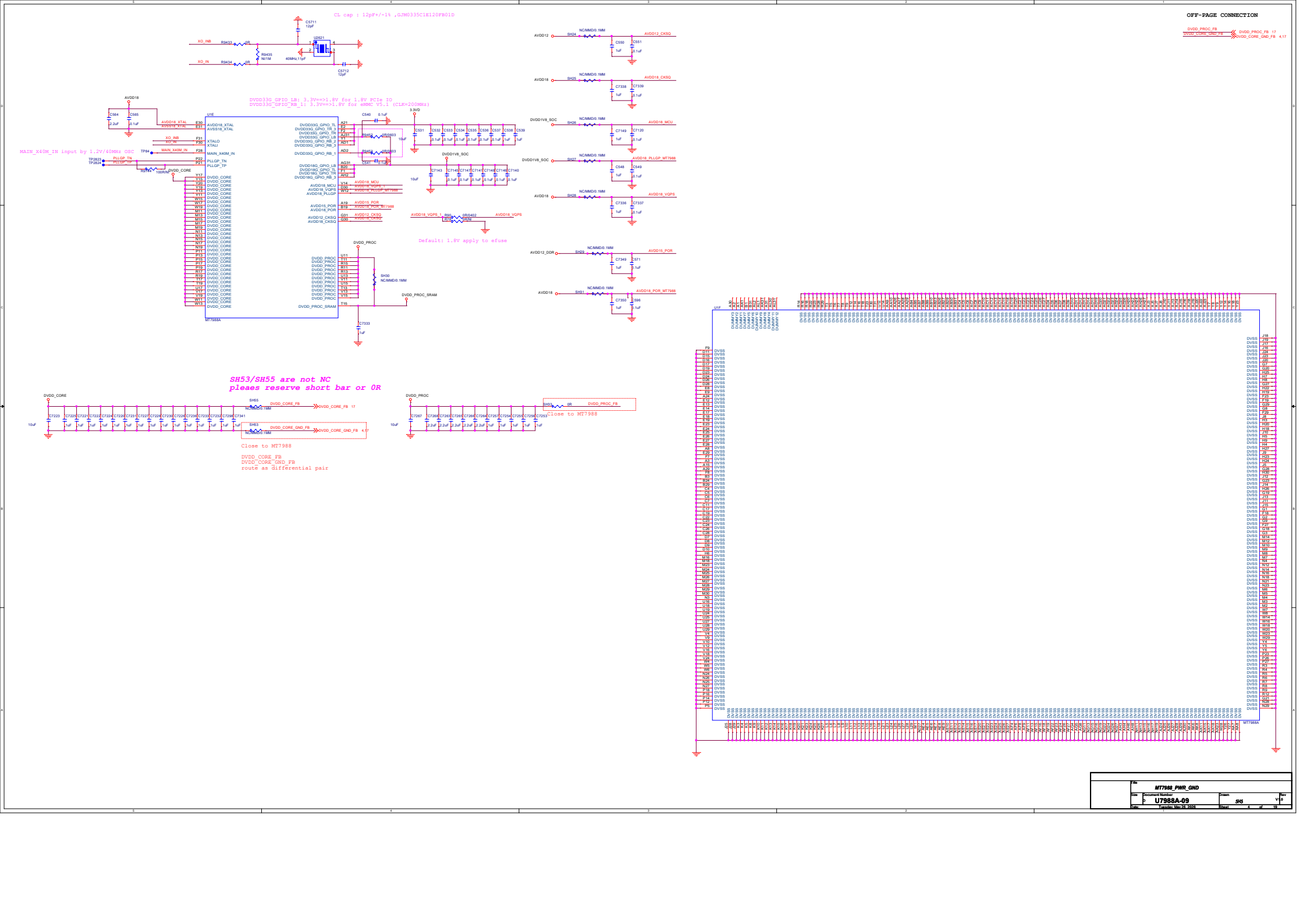
Title: MT7988_DRAM				
Size: C	Document Number: U7988A-09		Drawn	Rev: V1.0
Date: Tuesday, May 26, 2026			Sheet 2 of 19	



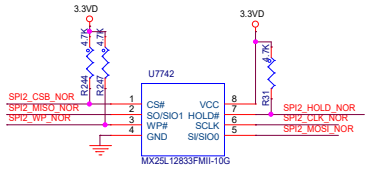
Function	Signal	Don't be used
PCIe	TX/RX/CLK pairs	Floating
	PERSET, CLKREQ, Wake	Floating
USXGMII/HSGMII	TX/RX pairs	Floating
USB3	TX/RX pairs	Floating
USB2	DP/DM	Floating
1GPHY x 4p	A/B/C/D pairs, REXT	Floating
2.5GPHY x 1p (Ready in 2023/Q3)	A/B/C/D pairs	Floating

Keep Serdes and GPHY signals floating (don't be used)

Title	MT7988_PCIe_USB_USXGMII		
Size	Document Number	Drawn	Rev
D	U7988A-09		V1.0
Date:	Tuesday, May 26, 2026	Sheet	3 of 19

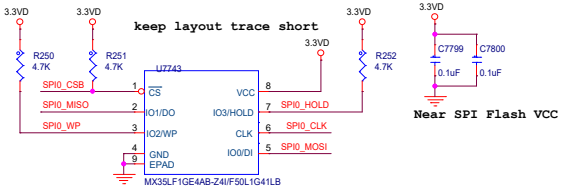

CL cap : 12pF+/-1% ,GJM0335C1E120FB01D
C5711 12pF
XO_INB
R9433 0R
XO_IN
R9434 0R
R9435 NI/1M
U2621
40MHz,11pF
C5712 12pF
AVDD18
C564 2.2uF
C565 0.1uF
AVDD18_XTAL
AVSS18_XTAL
XO_INB
XO_IN
MAIN_X40M_IN input by 1.2V/40MHz OSC
TP84
MAIN_X40M_IN
TP2822
TP2823
PLLGP_TN
PLLGP_TP
R9444 100R/NP
DVDD_CORE
U1E
MT7988A
DVDD33G_GPIO_LB: 3.3V==>1.8V for 1.8V PCIe IO
DVDD33G_GPIO_RB_1: 3.3V==>1.8V for eMMC V5.1 (CLK=200MHz)
3.3VD
C540 0.1uF
R9452 0R/0603
C541 0.1uF
R9453 0R/0603
10uF
C531 C532 C533 C534 C535 C536 C537 C538 C539
DVDD1V8_SOC
C7143 C7145 C7147 C7141 C7140 C7144 C7140
AVDD18_MCU
AVDD18_VQPS
AVDD18_PLLGP
AVDD15_POR
AVDD18_POR
AVDD12_CKSQ
AVDD18_CKSQ
AVDD18_VQPS_1
R9454 0R/0402
AVDD18_VQPS
Default: 1.8V apply to efuse
DVDD_PROC
SH30 NC/MMD/0.1MM
DVDD_PROC_SRAM
C7333 1uF
AVDD12
SH24 NC/MMD/0.1MM
C550 1uF
C551 0.1uF
AVDD12_CKSQ
AVDD18
SH25 NC/MMD/0.1MM
C7338 1uF
C7339 0.1uF
AVDD18_CKSQ
DVDD1V8_SOC
SH26 NC/MMD/0.1MM
C7149 1uF
C7120 0.1uF
AVDD18_MCU
DVDD1V8_SOC
SH27 NC/MMD/0.1MM
C548 1uF
C549 0.1uF
AVDD18_PLLGP_MT7988
AVDD18
SH28 NC/MMD/0.1MM
C7336 1uF
C7337 0.1uF
AVDD18_VQPS
AVDD12_DDR
SH29 NC/MMD/0.1MM
C7349 1uF
C571 0.1uF
AVDD15_POR
AVDD18
SH31 NC/MMD/0.1MM
C7350 1uF
C396 0.1uF
AVDD18_POR_MT7988
OFF-PAGE CONNECTION
DVDD_PROC_FB
DVDD_CORE_GND_FB
DVDD_PROC_FB 17
DVDD_CORE_GND_FB 4,17
SH53/SH55 are not NC
pleaes reserve short bar or 0R
DVDD_CORE
SH55 NC/MMD/0.1MM
DVDD_CORE_FB
DVDD_CORE_FB 17
SH53 NC/MMD/0.1MM
DVDD_CORE_GND_FB
DVDD_CORE_GND_FB 4,17
C7223 C7224 C7225 C7226 C7227 C7228 C7229 C7230 C7231 C7232 C7233 C7234 C7235 C7341
10uF
1uF
Close to MT7988
DVDD_CORE_FB
DVDD_CORE_GND_FB
route as differential pair
DVDD_PROC
SH53 0R
DVDD_PROC_FB
Close to MT7988
C7267 C7260 C7261 C7262 C7263 C7264 C7265 C7254 C7255 C7256 C7257
2.2uF
U1F
MT7988A
Title
MT7988_PWR_GND
Size
D
Document Number
U7988A-09
Drawn
SH5
Rev
V1.0
Date:
Tuesday, May 26, 2026
Sheet
4
of
19

SPI2 to SPIM-NOR Flash

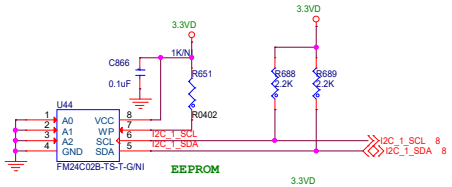


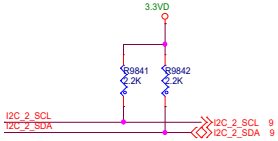
SPI Nor Flash

SPI0 to SPIM-NAND/SNFI-NAND Flash

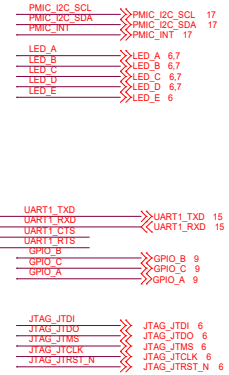


SPI NAND Flash **SPIM NAND**
ECC Free NAND





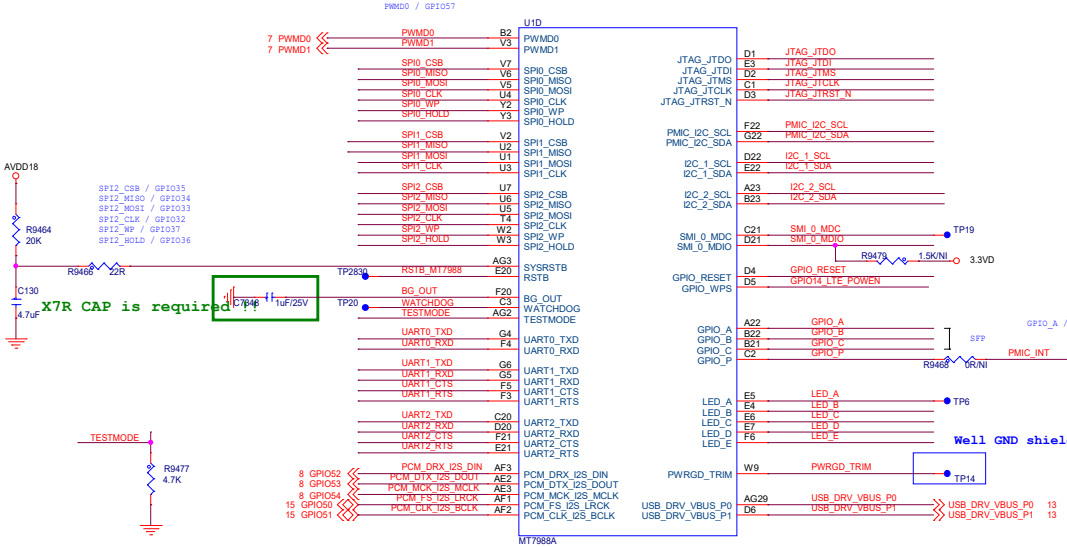
OFF-PAGE CONNECTION

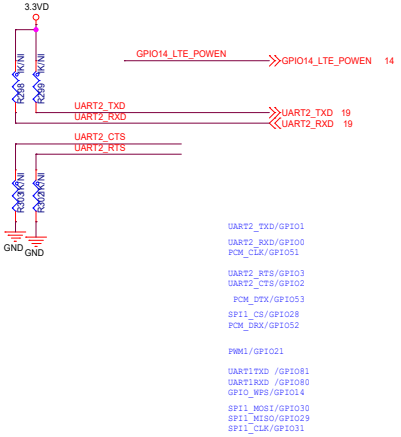


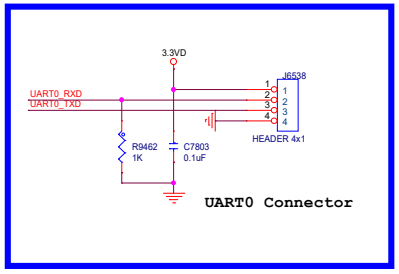
SPI2_CLK	B1:MSDC0_DAT3
SPI2_MOSI	B1:MSDC0_DAT2
SPI2_MISO	B1:MSDC0_DAT1
SPI2_CSB	B1:MSDC0_DAT0
SPI2_HOLD	B1:MSDC0_CMD
SPI2_WP	B1:MSDC0_CLK

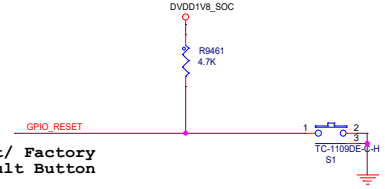
SPI2_CLK	SPI2_CLK_NOR
SPI2_MOSI	SPI2_MOSI_NOR
SPI2_MISO	SPI2_MISO_NOR
SPI2_CSB	SPI2_CSB_NOR
SPI2_HOLD	SPI2_HOLD_NOR
SPI2_WP	SPI2_WP_NOR

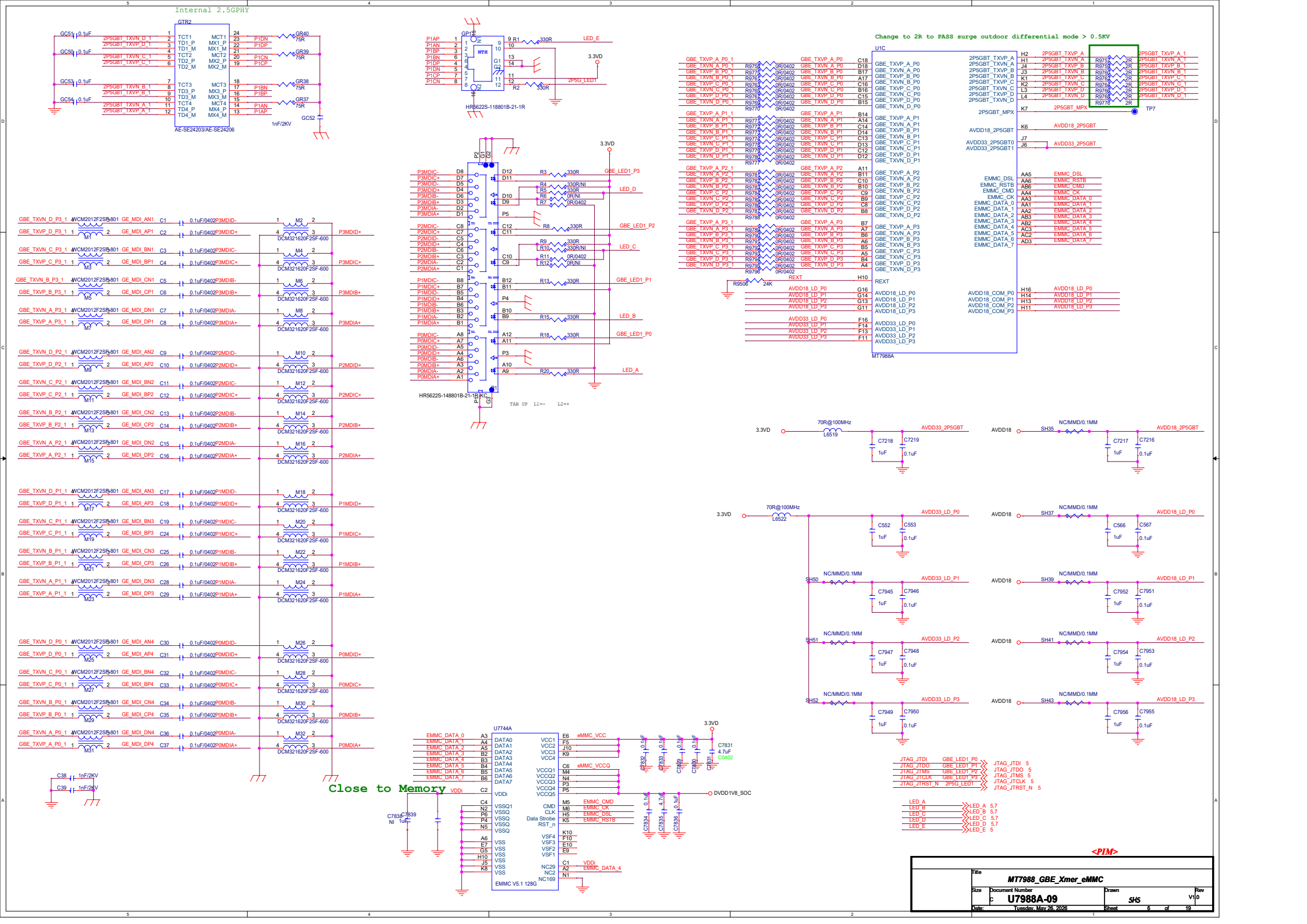
Reset Circuit

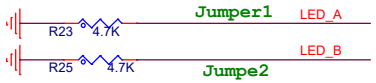


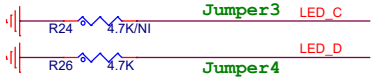







Internal 2.5GPHY
GTR2
AE-SE24203/AE-SE24206
Change to 2R to PASS surge outdoor differential mode > 0.5KV
U1C
MT7988A
HR5622S-118801B-21-1R
HR5622S-148801B-21-1R-KC
TAB UP L1=- L2=+
Close to Memory
U7744A
EMMC V5.1 128G
<PIM>
MT7988_GBE_Xmer_eMMC
U7988A-09
SH5
Tuesday, May 26, 2026
Sheet 6 of 19



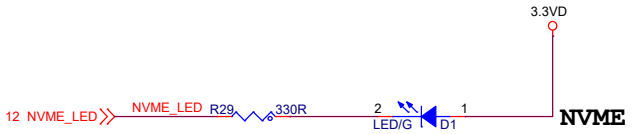


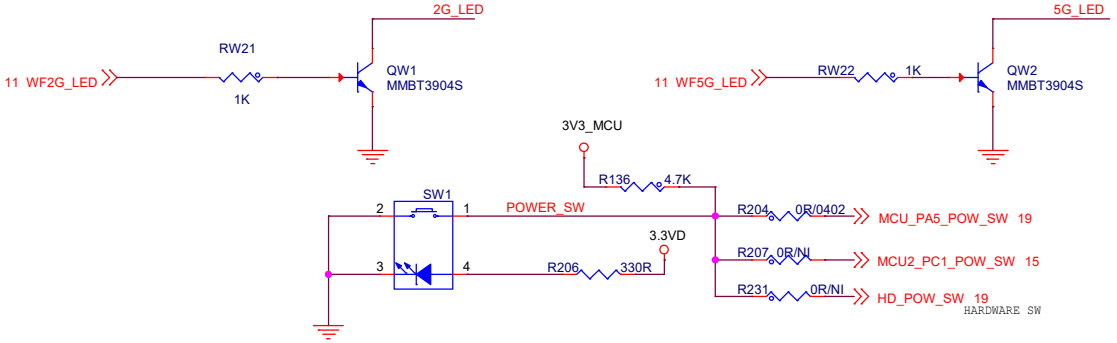
OFF-PAGE CONNECTION

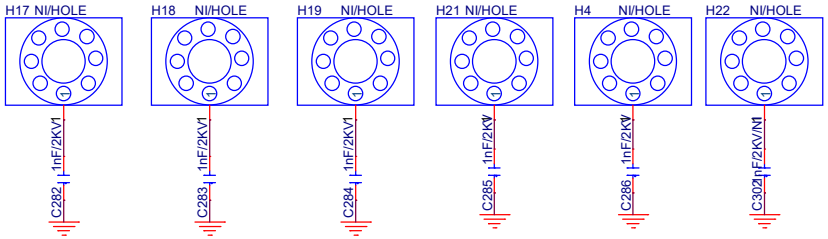


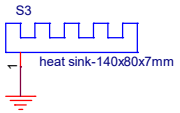
注意LED极性要跟着改

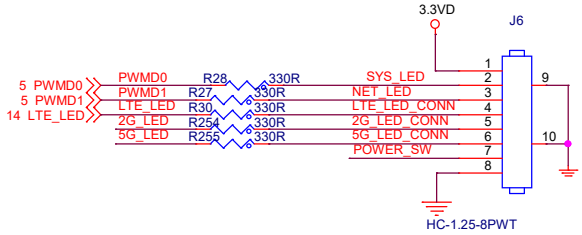
Boot Strapping Pin Name	Description	Boot Sequence
LED_A	Ref. CLK Source	0: By XTAL 1: By Oscillator
LED_B	DDR Type	0: DDRx16x2 1: DDRx16x1
(LED_D, LED_C)	Boot mode	00: SPI-NOR 01: SNAND(SPI_M)==>SD 10: eMMC 11: SNAND(SFNI)==>SD

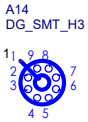


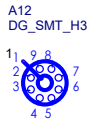


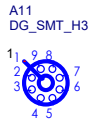


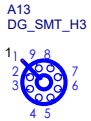














TOP SIDE FIDUCIALS

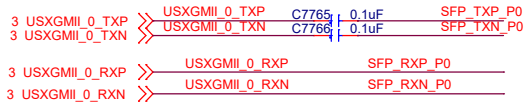


BOTTOM SIDE FIDUCIALS

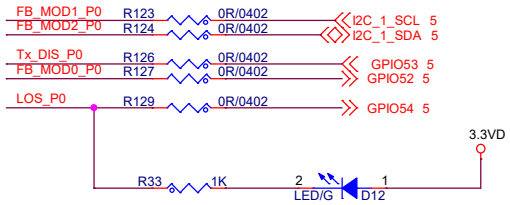


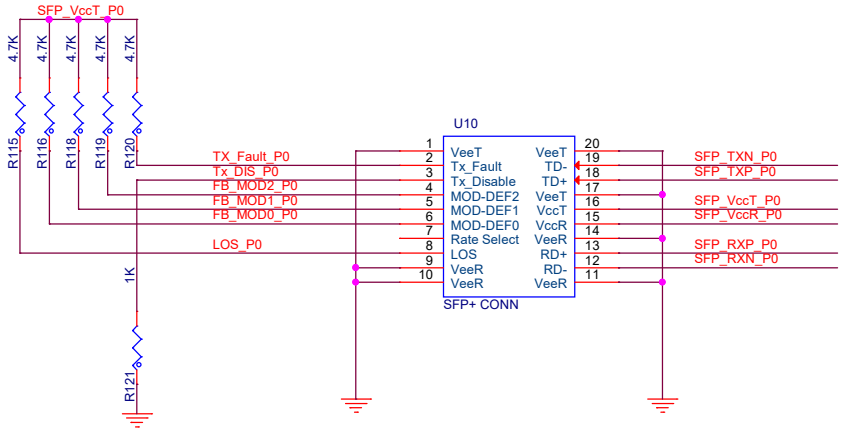
<PIM>

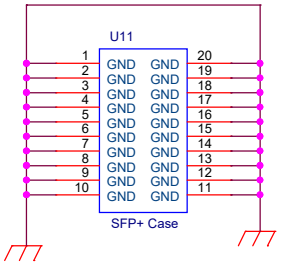
Title: MT7988_Boot Strap_LEDs			
Size: B	Document Number: U7988A-09	Drawn: SHS	Rev: V1.0
Date: Wednesday, June 03, 2026		Sheet 7 of 19	

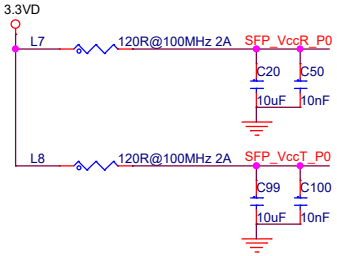


OFF-PAGE CONNECTION









Title: *USXGMII_P0_SFP+*			
Size: B	Document Number: **U7988A-09**	Drawn: *SHS*	Rev: V1.0
Date: Tuesday, May 26, 2026		Sheet 8 of 19	

Layout co-pad

3 USXGMII_1_TXP
3 USXGMII_1_TXN
C292 0.1uF SFP_TXP_P1
C293 0.1uF SFP_TXN_P1

Differential 90-ohm

3 USXGMII_1_RXP
3 USXGMII_1_RXN
SFP_RXP_P1
SFP_RXN_P1

OFF-PAGE CONNECTION

FB_MOD0_P1 R9717 0R/0402 GPIO_A 5
FB_MOD1_P1 R9726 0R/0402 I2C_2_SCL 5
FB_MOD2_P1 R9724 0R/0402 I2C_2_SDA 5
Tx_DIS_P1 R9716 0R/0402 GPIO_B 5
LOS_P1 R9719 0R/0402 GPIO_C 5

3.3VD
R35 1K 2 LED/G 1 D13

SFP_VccT_P1
4.7K R9443
4.7K R9446
4.7K R9442
4.7K R9444
4.7K R9441
1K R9440

U7740

1	VeeT	VeeT	20
2	Tx_Fault	TD-	19
3	Tx_Disable	TD+	18
4	MOD-DEF2	VeeT	17
5	MOD-DEF1	VccT	16
6	MOD-DEF0	VccR	15
7	Rate Select	VeeR	14
8	LOS	RD+	13
9	VeeR	RD-	12
10	VeeR	VeeR	11

SFP+ CONN

TX_Fault_P1
Tx_DIS_P1
FB_MOD2_P1
FB_MOD1_P1
FB_MOD0_P1
LOS_P1

SFP_TXN_P1
SFP_TXP_P1
SFP_VccT_P1
SFP_VccR_P1
SFP_RXP_P1
SFP_RXN_P1

U7741

1	GND	GND	20
2	GND	GND	19
3	GND	GND	18
4	GND	GND	17
5	GND	GND	16
6	GND	GND	15
7	GND	GND	14
8	GND	GND	13
9	GND	GND	12
10	GND	GND	11

SFP+ Case

3.3VD
L6508 120R@100MHz 2A SFP_VccR_P1
C7776 10uF C7775 10nF
L6505 120R@100MHz 2A SFP_VccT_P1
C7773 10uF C7774 10nF

<PIM>

Title	USXGMII_P1_SFP+		
Size	Document Number	Drawn	Rev
B	U7988A-09	SHS	V1.0
Date:	Tuesday, May 26, 2026	Sheet	9 of 19



Shielding frame Hole (TOP)

WF0_2G_iPAiLNA

Close notch filter

WF1_2G_iPAiLNA

Close notch filter

WF2_5G_iPAiLNA

5-STEP-LPF

WF3_5G_iPAiLNA

5-STEP-LPF

WF4_5G_iPAiLNA

5-STEP-LPF

MT7976C

trace width 8/4/8mil

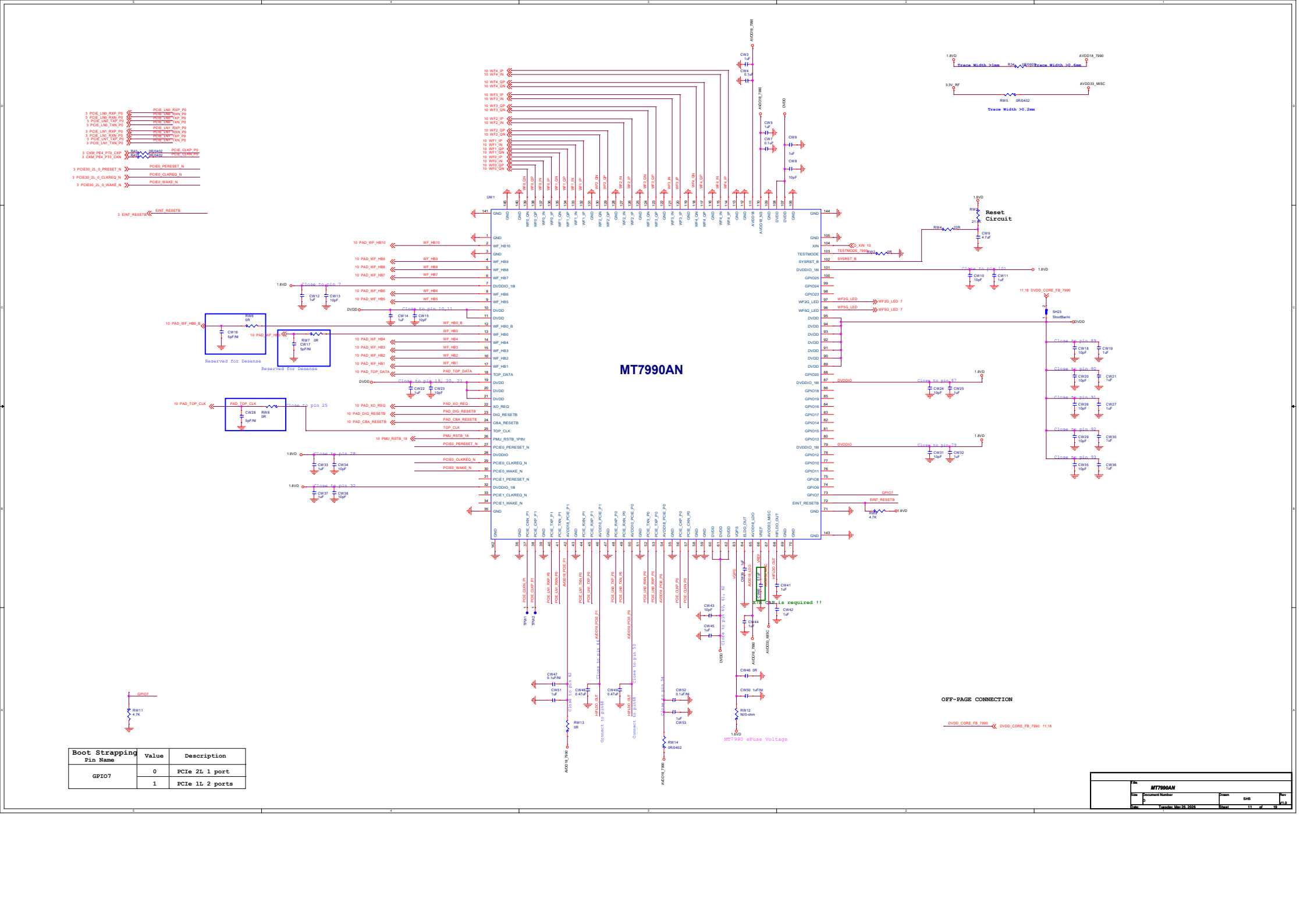
1. WF2/3/4 IQ Pairs w/solid GND plane and via
2. Trace width 4/4/4mil

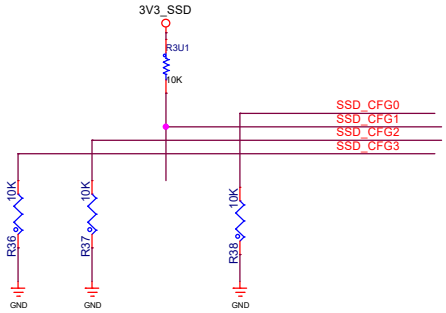
1. Trace width 4/4/4mil
2. Caps close to MT7976

For EMI reserved

<EMI>

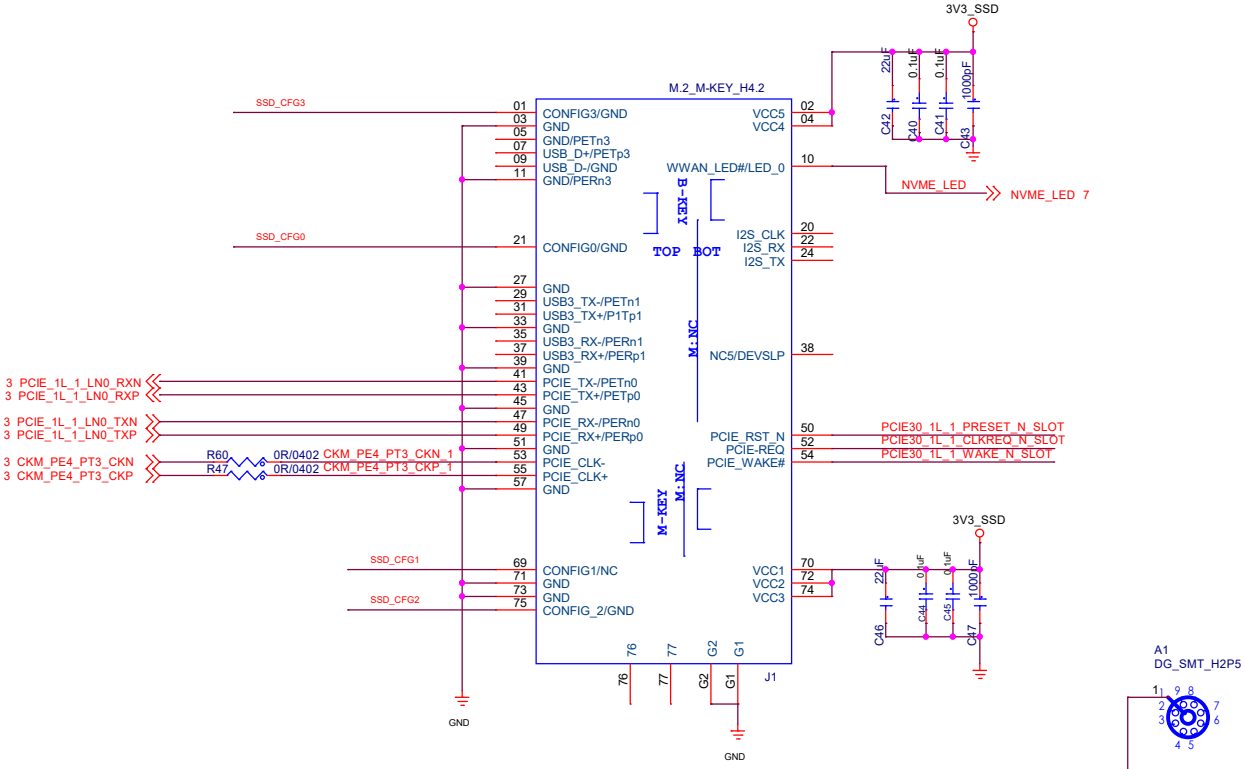
MT7976C

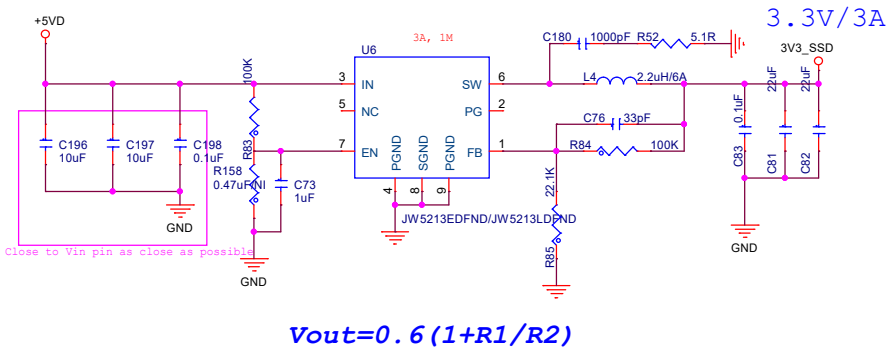

MT7990AN
Reset Circuit
OFF-PAGE CONNECTION
MT7990 eFuse Voltage
x7R Cap is required !!
Reserved for Desense
Reserved for Desense
Trace Width >1mm
Trace Width >0.6mm
Trace Width >0.2mm
Close to pin 7
Close to pin 10,11
Close to pin 19, 20, 21
Close to pin 25
Close to pin 28
Close to pin 32
Close to pin 101
Close to pin 87
Close to pin 79
Close to pin 83
Close to pin 90
Close to pin 91
Close to pin 92
Close to pin 93
Close to pin 42
Close to pin 50
Close to pin 54
Connect to plane
Connect to plane
Close to pin 45, 62
Close to pin 42
Close to pin 54
Close to pin 50
Close to pin 45, 62
Boot Strapping
Pin Name
Value
Description
GPIO7
0
PCIe 2L 1 port
1
PCIe 1L 2 ports
Title
MT7990AN
Size
D
Document Number
Drawn
SHS
Rev
V1.0
Date:
Tuesday, May 26, 2026
Sheet
11
of
19



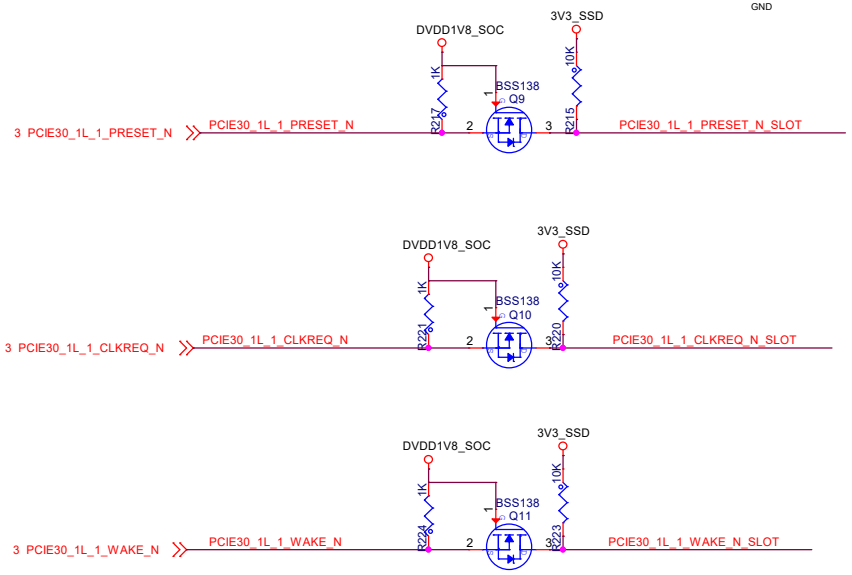
M2_WWAN_CONFIG

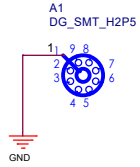
CFG0	CFG1	CFG2	CFG3	MODE
L	L	L	L	SSD - SATA
L	H	L	L	SSD - PCIE





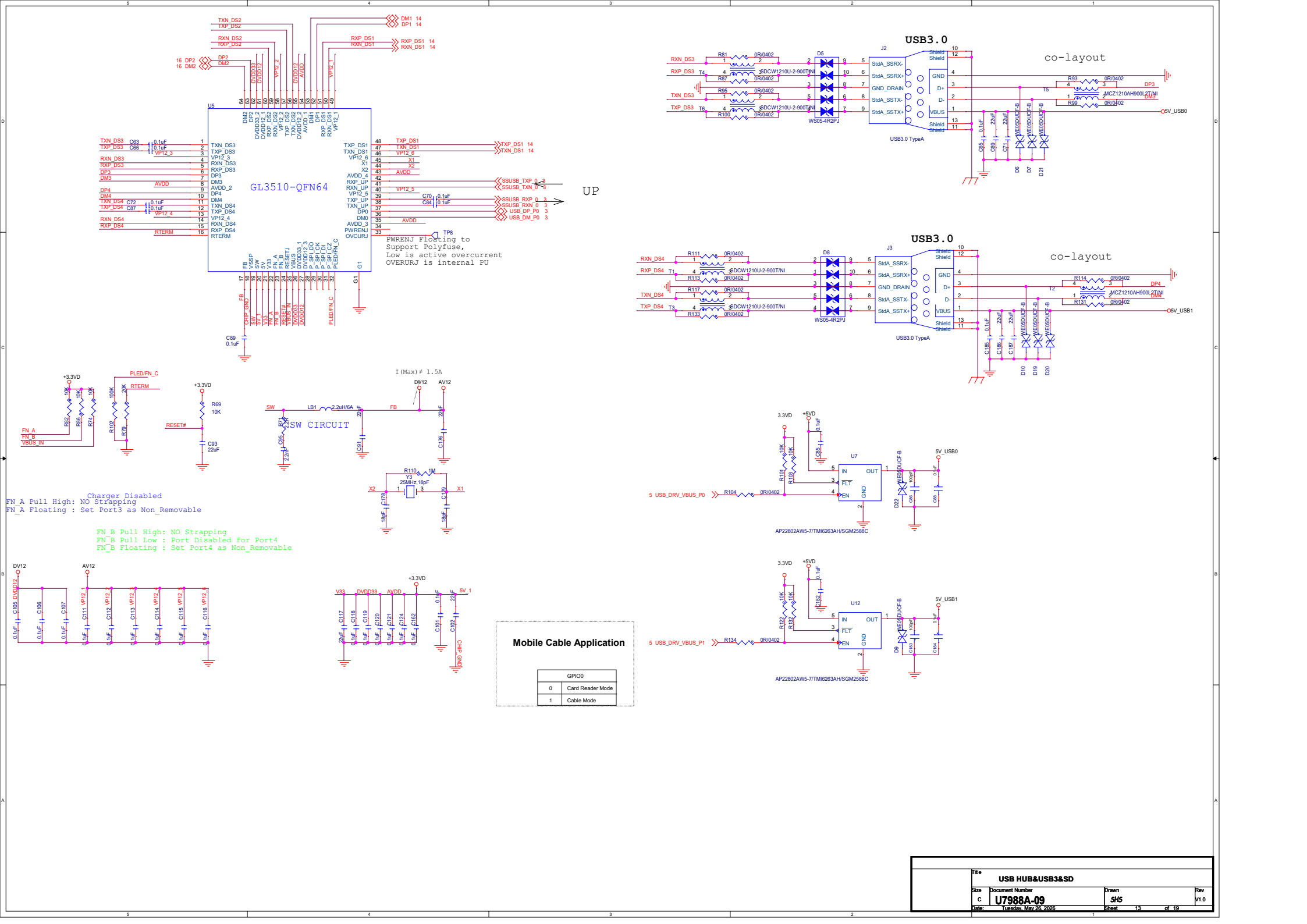
$$Vout=0.6(1+R1/R2)$$

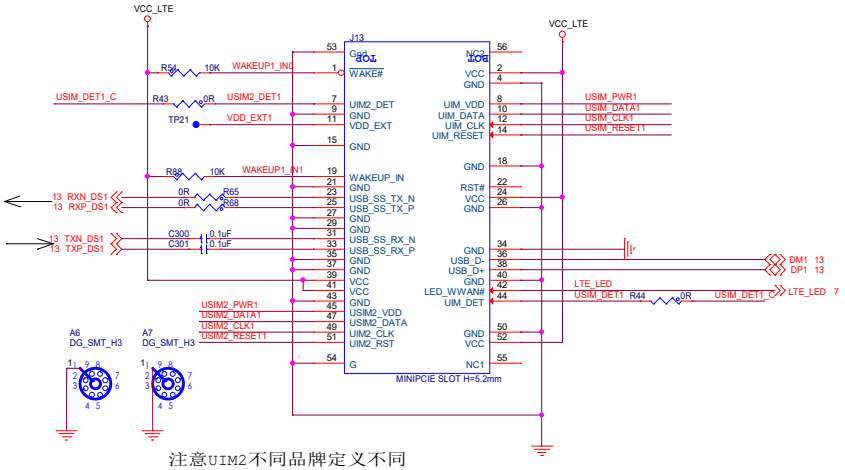




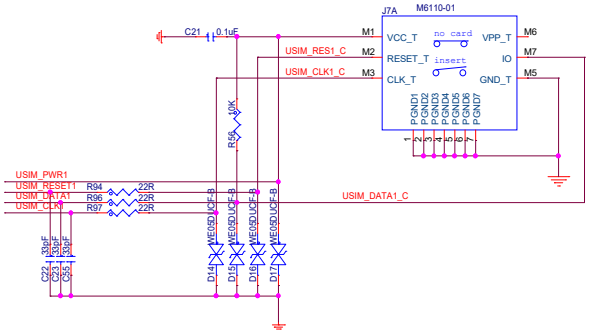
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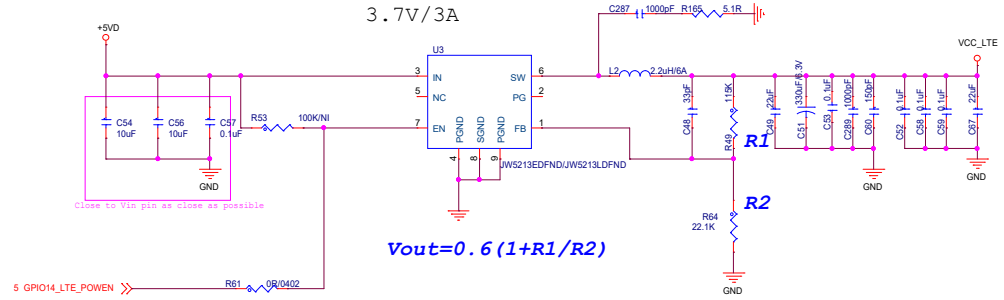
Title: SSD			
Size:	Document Number: A3 U7988A-09	Drawn: SHS	Rev: V1.0
Date: Tuesday, May 26, 2026		Sheet 12 of 19	

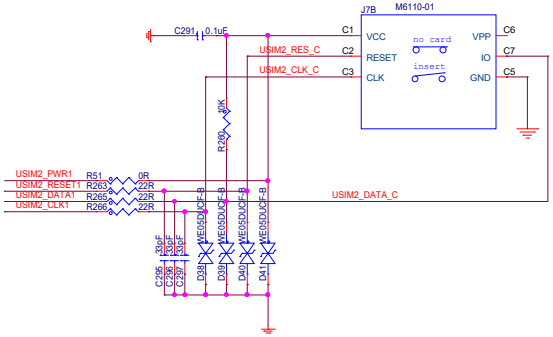

GL3510-QFN64
U5
UP
PWRENJ Floating to
Support Polyfuse,
Low is active overcurrent
OVERURJ is internal PU
TP8
USB3.0
co-layout
USB3.0 TypeA
WS05-4R2PJ
SDCW1210U-2-900T/NI
MCZ1210AH900L2T/NI
WE05DUCF-B
I(Max)≠ 1.5A
SW CIRCUIT
Charger Disabled
FN_A Pull High: NO Strapping
FN_A Floating : Set Port3 as Non_Removable
FN_B Pull High: NO Strapping
FN_B Pull Low : Port Disabled for Port4
FN_B Floating : Set Port4 as Non_Removable
AP22802AW5-7/TMI6263AH/SGM2588C
Mobile Cable Application
GPIO0
0 Card Reader Mode
1 Cable Mode
USB HUB&USB3&SD
U7988A-09
Tuesday, May 26, 2026
SHS
V1.0
13 of 19

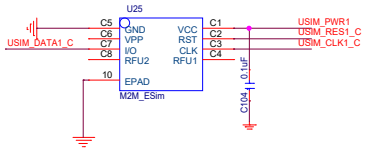


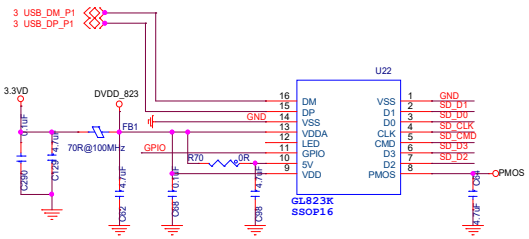
注意UIM2不同品牌定义不同

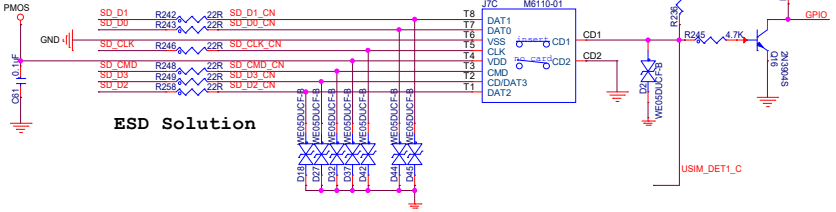


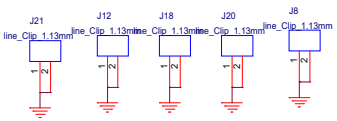




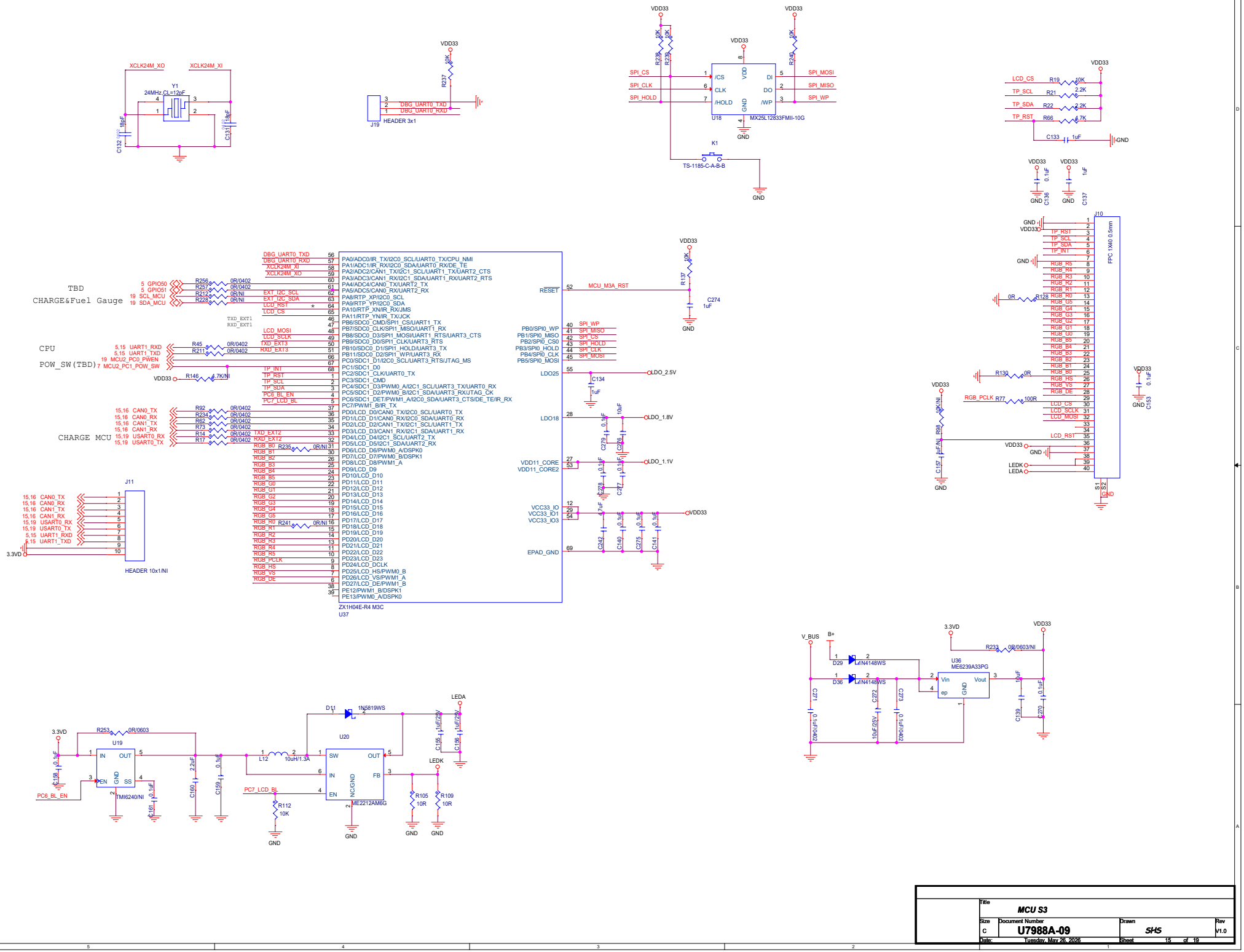




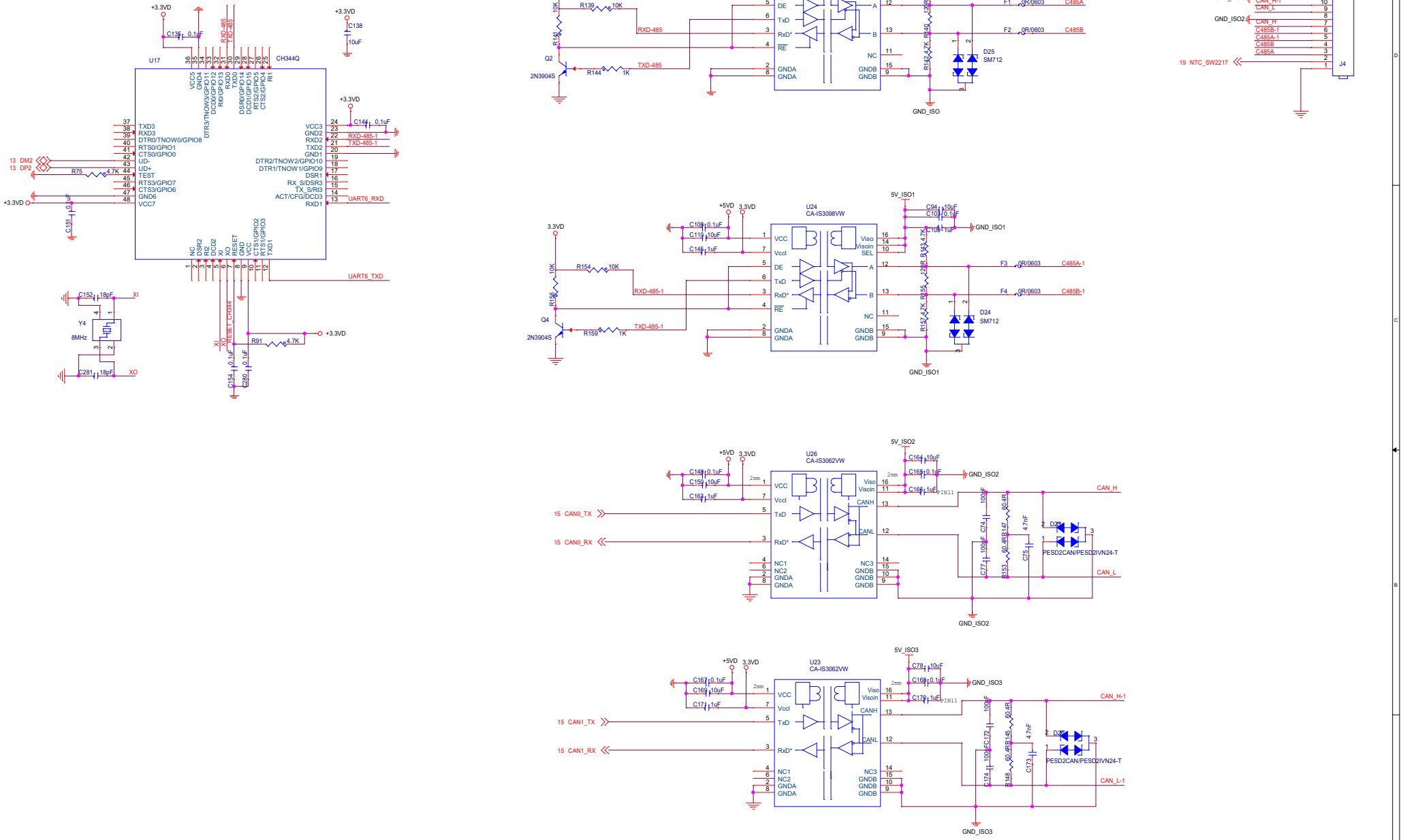




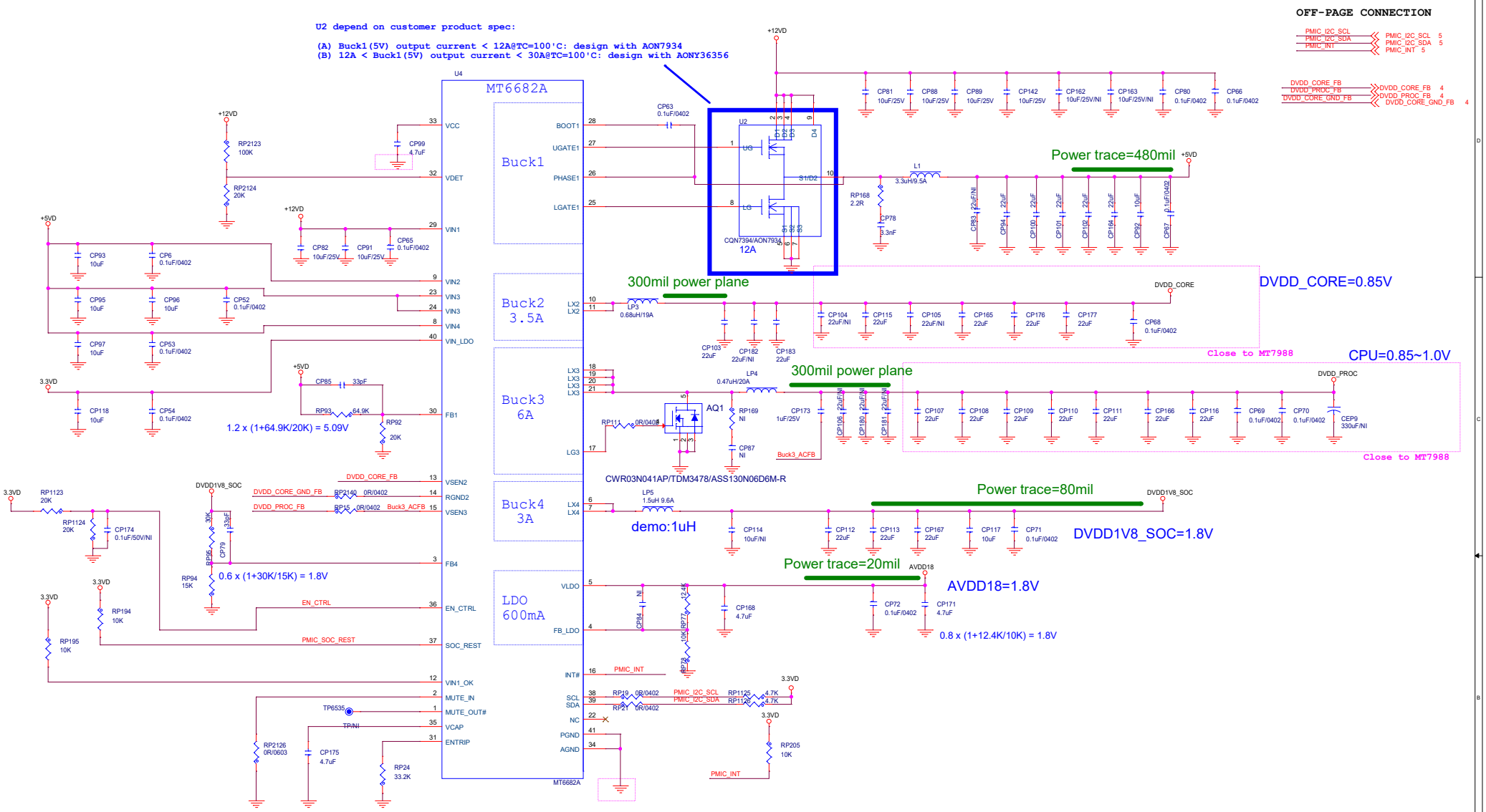
Title: LTE&SD			
Size: C	Document Number: U7988A-09	Drawn: SHS	Rev: V1.0
Date: Tuesday, May 26, 2026		Sheet 14 of 19	



Title: MCU S3			
Size: C	Document Number: U7988A-09	Drawn: SHS	Rev: V1.0
Date: Tuesday, May 26, 2026		Sheet 15 of 19	



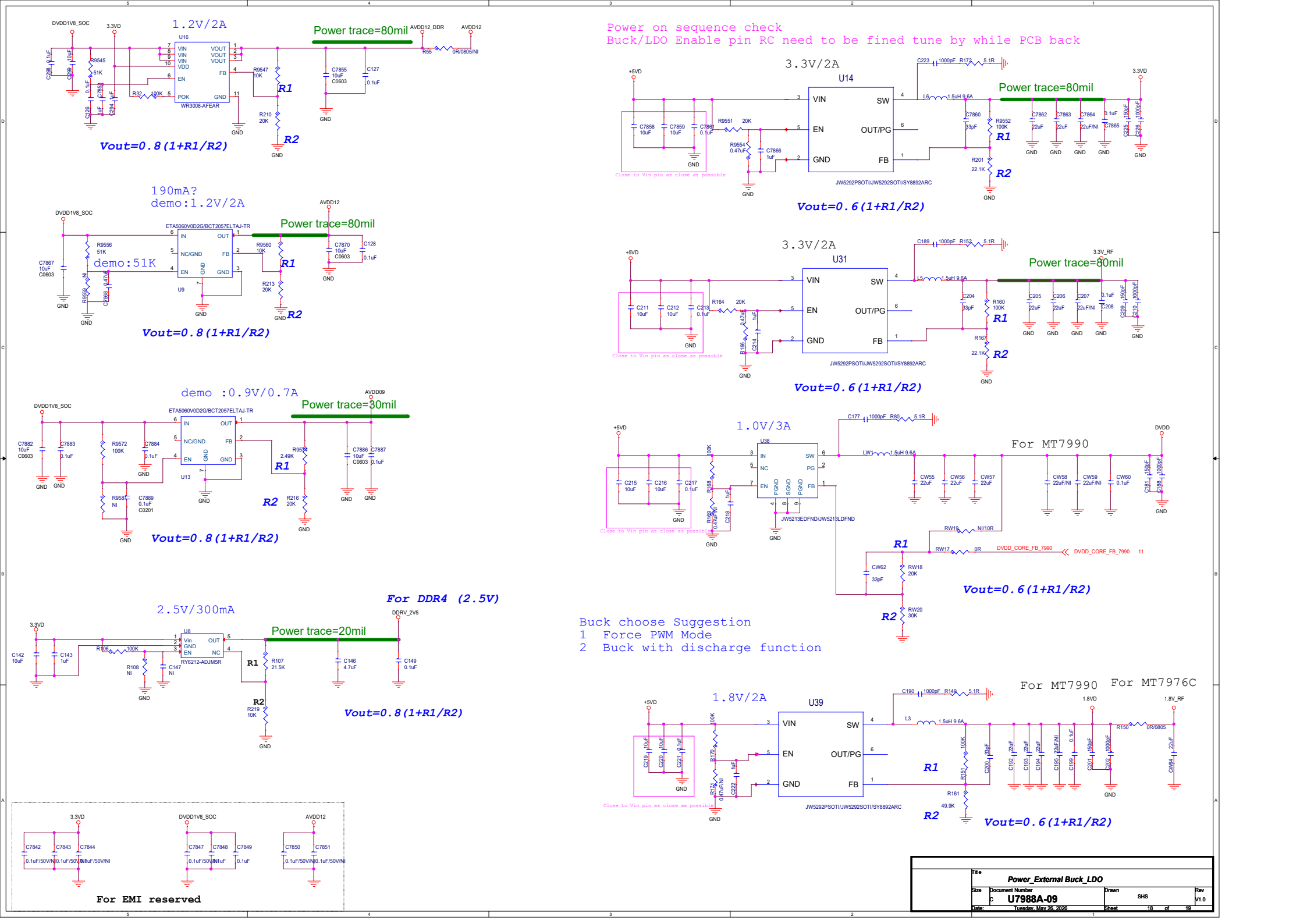
Title	485 & CAN & Conn		
Size: C	Document Number: U7988A-09	Drawn: SHS	Rev: V1.0
Date: Tuesday, May 26, 2026		Sheet 16 of 19	

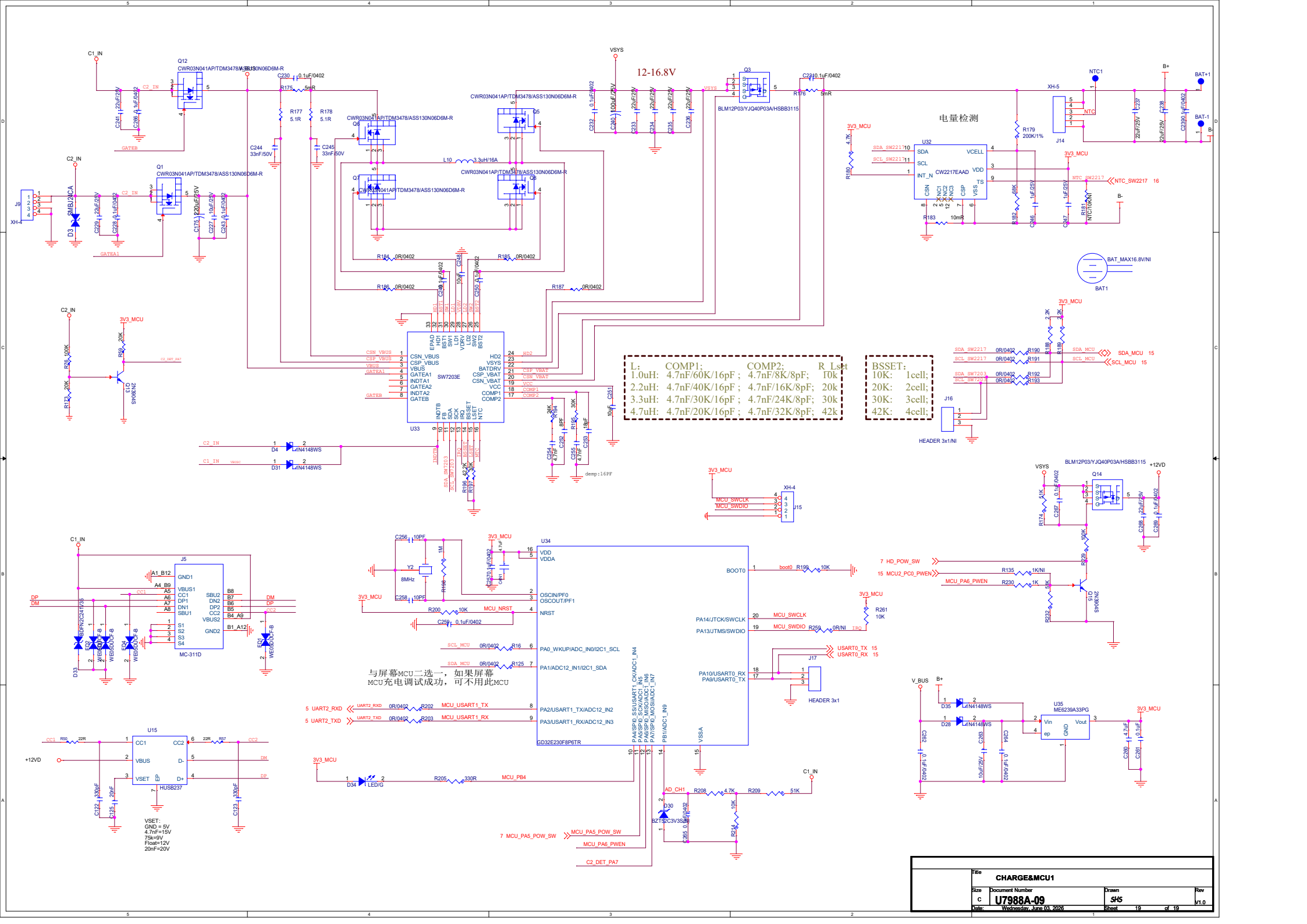


RT5190C HV/NV/LV Jumper

Pin Name	RT5190C All output power
(CLKREQ,_N WAKE_N)	00: All LV 11: ALL HV 01: ALL NV 10: ALL NV

Title: PMIC_MT6682A			
Size: C	Document Number: U7988A-09	Drawn	Rev: V1.0
Date: Tuesday, May 26, 2026	Sheet 17 of 19		


1.2V/2A
Power trace=80mil
Vout=0.8(1+R1/R2)
190mA?
demo:1.2V/2A
demo:51K
Vout=0.8(1+R1/R2)
demo :0.9V/0.7A
Power trace=30mil
Vout=0.8(1+R1/R2)
For DDR4 (2.5V)
2.5V/300mA
Power trace=20mil
Vout=0.8(1+R1/R2)
For EMI reserved
Power on sequence check
Buck/LDO Enable pin RC need to be fined tune by while PCB back
3.3V/2A
Power trace=80mil
Vout=0.6(1+R1/R2)
3.3V/2A
Power trace=80mil
Vout=0.6(1+R1/R2)
1.0V/3A
For MT7990
Vout=0.6(1+R1/R2)
Buck choose Suggestion
1 Force PWM Mode
2 Buck with discharge function
1.8V/2A
For MT7990
For MT7976C
Vout=0.6(1+R1/R2)
Power_External Buck_LDO
U7988A-09
SHS
V1.0
Tuesday, May 26, 2026
18
19


12-16.8V
电量检测
L: COMP1: COMP2: R_Lset
1.0uH: 4.7nF/60K/16pF ; 4.7nF/8K/8pF; 10k
2.2uH: 4.7nF/40K/16pF ; 4.7nF/16K/8pF; 20k
3.3uH: 4.7nF/30K/16pF ; 4.7nF/24K/8pF; 30k
4.7uH: 4.7nF/20K/16pF ; 4.7nF/32K/8pF; 42k
BSSET:
10K: 1cell;
20K: 2cell;
30K: 3cell;
42K: 4cell;
SW7203E
GD32E230F8P6TR
SW7220-3
CW2217EAAD
BLM12P03/YJQ40P03A/HSBB3115
CWR03N041AP/TDM3478/ASS130N06D6M-R
MC-311D
HUSB237
BAT_MAX16.8V/NI
HEADER 3x1/NI
HEADER 3x1
与屏幕MCU二选一，如果屏幕MCU充电调试成功，可不用此MCU
VSET:
GND = 5V
4.7nF=15V
75k=9V
Float=12V
20nF=20V
CHARGE&MCU1
U7988A-09
V1.0
Wednesday, June 03, 2026